

WEIWEN YUAN

PhD Student | School of Science and Engineering

The Chinese University of Hong Kong, Shenzhen

Email: 223010145@link.cuhk.edu.cn |

Google Scholar | <https://scholar.google.com/citations?user=hOG8ui0AAAAJ>

Personal Website | <https://yuanweiwei666.github.io/YuanWeiwen.github.io/>

Expected graduation: May 2027 | Available from July 2027

RESEARCH PROFILE

My research focuses on multi-task semantic communication for wireless image, speech, and video transmission. My first-author work combines generative models, large language models (LLMs), and intelligent agents to develop task-oriented communication systems. My future research interests lie in cloud-edge-device collaborative semantic communication, with the goal of integrating LLM-enabled agents more deeply into semantic communication systems. I am particularly interested in developing agentic SemCom architectures that can coordinate semantic understanding, task reasoning, and adaptive transmission across the cloud, edge, and devices, while responding to changing channel conditions, task requirements, and network resources.

EDUCATION

PhD in Computer and Information Engineering (in progress) | Sep 2023 – Present

The Chinese University of Hong Kong, Shenzhen | School of Science and Engineering

Supervisor: Prof. Shuguang Cui | Expected completion: May 2027

Master of Science in Electronic Engineering | Sep 2022 – Jul 2023

The Chinese University of Hong Kong

Bachelor's Degree in Electronic Engineering | Sep 2018 – Jun 2022

China University of Mining and Technology

DOCTORAL RESEARCH EXPERIENCE

The Chinese University of Hong Kong, Shenzhen | Sep 2023 – Present | Supervisor: Prof. Shuguang Cui

Conducted the experiments and drafted the manuscripts for four first-author journal and conference papers. [J1, J2, C1, C2]

FIRST-AUTHOR JOURNAL RESEARCH

LISA for Multi-Task Image Semantic Communication

- Investigated and evaluated LISA, a unified semantic-agent framework for image reconstruction, style transfer, and classification. Its semantic module combines multi-source prompting, an LLM, memory, and auxiliary tools to derive task-adaptive representations from images, task requirements, and environmental information. [J1]
- The channel module combines vector-quantized variational autoencoding, a shared codebook, and transformer-based modulation for digital semantic transmission. Experiments showed better performance than state-of-the-art baselines across all three tasks with reduced communication overhead.

AudioLLM-Enabled Multi-Task Speech Semantic Communication

- Investigated and evaluated a unified generative framework for wireless speech reconstruction, voice conversion, automatic speech recognition, and speech-to-text translation. Residual vector quantization encodes hierarchical speech factors into compact semantic codewords using shared codebooks. [J2]
- At the receiver, an RVQ decoder recovers semantic representations; an AudioLLM extracts multi-level semantic factors for lightweight task heads. Simulations showed consistent gains over baselines across all four tasks under AWGN and Rayleigh fading channels.

COLLABORATIVE JOURNAL SURVEYS AND TUTORIALS

Generative Semantic Communication Architectures and Technologies

- Conducted literature research and contributed to the initial manuscript of a survey covering VAE-, GAN-, diffusion-, and LLM-based semantic communication, including their architectures, applications, and open research questions. [J4]
- The survey also introduces LLM-based agents at the transmitter and receiver for semantic understanding and content regeneration, illustrating a shift from information recovery to information regeneration.

Retrieval-Augmented LLMs for Mobile Edge Computing

- Took primary responsibility for the literature review and drafting of Section 2 of a tutorial on RAG-enabled mobile edge computing, organizing and synthesizing relevant prior work. [J3]
- The tutorial examines how retrieved domain knowledge supports LLM decision-making at the edge, discussing method trade-offs, failure modes, and open challenges, with multi-user offloading as an illustrative application.

CONFERENCE AND WORKSHOP RESEARCH

Large Semantic Agents for Wireless Image Transmission

- Investigated and evaluated paired LLM-based semantic agents serving as transmitter and receiver codecs for point-to-point wireless image transmission. Each agent combines multi-source prompting, a pretrained LLM, and auxiliary tools such as APIs and databases. [C1]
- The agents perform semantic understanding and embedding generation at the transmitter, and content regeneration and image optimization at the receiver. Experiments showed better image transmission performance than four baseline schemes.

Generative Semantic Communication for Image Transmission and Segmentation

- Investigated and evaluated a multi-task system combining source and task knowledge bases with semantic-similarity-based instruction mapping to select task-relevant image features. [C2]
- The architecture uses a hierarchical Swin Transformer, a unified residual-block JSCC encoder, and task-specific decoders, including a diffusion decoder for reconstruction. Experiments showed improved PSNR and segmentation accuracy over prior single-task systems.

UNPUBLISHED AND ONGOING RESEARCH

VINTA for Task-Adaptive Video Semantic Communication

- Investigated VINTA, a cognitive-agent framework combining task-aware keyframe selection, LLM-based chain-of-thought reasoning, and tool-augmented actions for video reconstruction and question answering. [W1]
- The study introduces a multi-task video dataset for task-specific reasoning and reports improved transmission efficiency and robustness over baselines across communication conditions. Unpublished manuscript.

Edge-Coordinated Instruction-Level Self-Evolution for Video Question Answering

- Investigating ILSE, an edge-assisted framework that jointly adapts transmitter and receiver instructions without updating client model parameters. Receiver generation confidence triggers edge assistance; an edge agent retrieves, evaluates, and, when needed, regenerates task- and channel-matched instruction strategies. [U1]
- On TGIF-QA, ILSE recovered BERTScore-F1 and AlignScore faster than prompt-adaptation baselines after two scheduled changes in task type and channel conditions, with an average downlink overhead of 75.69 bytes per evolution event. Under review at IEEE ICC.

Edge-Client Collaborative Dual-Level Adaptation

- Investigating edge-side prompt adaptation and model-parameter fine-tuning to update client-side transmitter and receiver agents as channels and task requirements change, extending the instruction-level adaptation direction to task-oriented video semantic communication. Manuscript in preparation. [M1]

PUBLICATIONS

First-Author Journal Articles

[J1] **W. Yuan**, J. Ren, X. Zhang, T. Jiang, Y. Zhang, R. Sun, and S. Cui, "Semantic-Aware Intelligent Agent for Multi-Task Image Communications," *IEEE Transactions on Cognitive Communications and Networking*, 2026.

[J2] **W. Yuan**, J. Zheng, J. Ren, X. Zhang, Y. Shen, Z. Xing, S. Wang, and S. Cui, "AudioLLM-Enabled Semantic Communication for Multi-Task Speech Transmission," *IEEE Transactions on Consumer Electronics*, 2026.

[J3] R. Ren[†], **W. Yuan**[†], T. Jiang, X. Zhang, Y. Wu, J. Ren, Z. Xing, C. Wang, H. Xing, Y. Shen, S. Wang, and K. F. Tsang, "Mobile Edge Computing with Retrieval-Augmented Large Language Models: A Tutorial," *IEEE Transactions on Consumer Electronics*, 2026. (†co-first authors)

Collaborative Journal Surveys and Tutorials

[J4] J. Ren, Y. Sun, H. Du, **W. Yuan**, C. Wang, X. Wang, Y. Zhou, Z. Zhu, F. Wang, and S. Cui, "Generative Semantic Communication: Architectures, Technologies, and Applications," *Engineering*, vol. 56, pp. 45–61, 2026.

Conference and Workshop Papers

[C1] **W. Yuan**, J. Ren, R. Sun, Y. Han, and S. Cui, "Large Semantic Agents for Wireless Image Transmission," *2025 IEEE 26th International Workshop on Signal Processing and Artificial Intelligence for Wireless Communications (SPAWC)*, pp. 1–5, 2025.

[C2] **W. Yuan**, J. Ren, C. Wang, R. Zhang, J. Wei, D. I. Kim, and S. Cui, "Generative Semantic Communication for Joint Image Transmission and Segmentation," *2025 IEEE International Conference on Communications Workshops (ICC Workshops)*, pp. 1110–1115, 2025.

MANUSCRIPT UNDER REVIEW

[U1] "Edge-Coordinated Self-Evolving Semantic Communication for Video Question Answering." Under review at IEEE ICC.

MANUSCRIPT IN PREPARATION

[M1] "Edge-Client Collaborative Dual-Level Adaptation for Task-Oriented Video Semantic Communication." Manuscript in preparation.

ADDITIONAL UNPUBLISHED WORK

[W1] "Think When You Transmit: A Cognitive Agent for Task-Adaptive Video Communication." Unpublished manuscript.

EARLIER RESEARCH EXPERIENCE

Master's Research Project | The Chinese University of Hong Kong | Sep 2022 – May 2023

End-to-End Photonics–Algorithm Co-design for Advanced Imaging Processing

- Investigated metalens arrays for three-dimensional high-dynamic-range imaging; designed a repeating 2×2 subarray with different focusing efficiencies and studied factors affecting imaging.

TECHNICAL SKILLS

Programming and computing: Python, MATLAB, C++, Linux.

SELECTED HONORS

First-Class Scholarship, China University of Mining and Technology. [Sep 2021]

Second Prize, 15th May Day Mathematical Modeling Competition. [Sep 2020]